LUSAELIMU.COM

THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MPWAPWA DISTRICT COUNCIL

FORM TWO PRE MOCK II EXAMINATION
033 – BIOLOGY

TIME: 2:00 HRS **MARCH 2026**

INSTRUCTIONS

1. This paper consists of three (3) sections **A, B,** and **C** with a total of ten (10) questions.
2. Attempt **ALL** questions.
3. All answers must be written in the spaces provided.
4. Read each question carefully before answering it**.**
5. All writing must be in blue or black ink EXCEPT for diagrams which must be drawn inpencil and labeled in ink.
6. Write your NAME on the top right-hand corner of every page in this question paper.

LUSAELIMU.COM

SECTION A (15 Marks)
Answer all questions in this section.

1. For each of the items (i) - (x), choose the correct answer from among the given alternatives and write its letter in the box provided.

i. Which nutritional disorder would result if a person consumes larger quantities of carbohydrates than body's requirement? A. Rickets B. Marasmus C. Beriberi D. Obesity

ii. The process by which green plants and other organisms use sunlight energy to manufacture their own food is known as;
A. Respiration B. Photosynthesis C. Digestion D. Nutrition

iii. Nutritional disorders are common problem associated with poor balanced diet mostly to children. Which among the following in not a nutritional disorder?
A. Anaemia B. Asthma C. Goiter D. Night blindness

iv. Which organisms belongs to Kingdom Plantae?
A. Fern and amoeba B. Moss and Mucor C. Fern and moss D. Liverwort and yeast

v. Which of the following groups of organisms belong to the same phylum
A. liver fluke, hook worm, earthworm, and spider B. tapeworm, earth worm, grasshopper and rat
C. butterfly, grasshopper, frog, and elephant D. crabs, grasshopper, spider, and centipede

vi. Which of the following set of structures represents organs?
A. Heart, Liver and Eye B. Heart, Blood and Eye
C. Brain, Skin and Blood D. Muscle, Liver and Shoot

vii. Hance is a carpenter and Candy is a farmer. A diet taken by Hance and Candy after their activities should contain large amount of;

A. Carbohydrates B. Protein C. Vitamin D. Fats and oils

viii. Two students Miranda and Careen were assigned by Sir. Obuntu to collect houseflies, grasshoppers, butterflies and bees for their next biology study. Which branch of biology were they preparing for study?

A. Anatomy B. Mycology C. Entomology D. Parasitology

ix. Ilankunda was doing an experiment to observe a plant cell using a compound microscope. If the microscope she was using had a Magnification power of the eyepiece lens of *X*10 and that of the high-power objective lens was *X*40. What would be the magnification of specimen under observation?

A. *X*10 B. *X*500 C. *X*400 D. *X*50

x. How would you test for 2mls of starch solution put in the test tube?

A. Adding 2mls of iodine solution and shake the mixture
B. Adding 2 drops of Sudan III reagent and shake thoroughly
C. Adding 2 drops of iodine solution and shake the mixture
D. Adding 2 drops of Benedict's solution and gently heating the mixture

ANSWERS

i)	ii)	iii)	iv)	v)	vi)	vii)	viii)	ix)	x)

2. Match the description in List A with the respective items in List B by writing the letter of the correct response in the boxes provided

LIST A	LIST B
i. The feeding relationship between two living organisms of different species in which one species benefit while the other neither benefits nor is harmed. ii. It is the type of heterotrophic nutrition whereby the organism feeds on complex food substances, digests, absorbs and assimilates/uses the nutrients. iii.The feeding relationship between two living organisms of different species in which both benefits. iv.The feeding relationship between two different species where one specie benefits while the other is harmed. v.The feeding relationship in which an organism feeds on dead or decaying bodies of other organisms, mainly plants and animals	A. Mutualism B. Endoparasites C. Amensalism D. Saprophytic nutrition E. Competition F. Parasitism G. Predation H. Commensalism I. Holozoic nutrition

ANSWERS

LIST A:	i)	ii)	iii)	iv)	v)
LIST B:					

3. (a) A mother of two kids took her kids to the hospital for medical examination, after examination of the temperature of the kids at Makole Hospital, the readings were recorded as shown below. Convert the temperatures recorded into degree Fahrenheit.

(i) First kid had 25^0C
(ii) Second kid had 273.15K

(b) The biology laboratory is different from other school facilities such as classroom, library as well as Physics and Chemistry laboratories. What distinguishes Biology Laboratory from Other School Facilities. Give four (4) points.

(i) ……

(ii) ……

(ii)……

(iii)……..

(iv)……

4. Form Two students classified moss and fern into the same Division simply because they have chlorophyll for photosynthesis.
   (a) Which classification system did they use in their process?……
   (b) Briefly explain four demerits of the classification system used in 4(a) above.
      (i) ……..…
      (ii) ……
      (iii) ……..……
      (iv) ……..……
5. Form II students performed an experiment to determine food substance contained in sample **Q.** They took 2mls of sample **Q** into a test tube. Then they added 2mls of sodium hydroxide followed by three drops of 1% Copper (II) Sulphate solution. A purple colour was observed.
   (a) Identify a food substance present in the solution **Q**……..…..……
   (b) Give three natural sources of food substance contained in the solution **Q**.
      (i)……...
      (ii) ……..……...
      (iii) ……..……...
   (c) State two functions of the food substance present in the solution **Q**.
      (i) ……...
      (ii) ……...
   (d) Name the deficiency that may arise when the body does not have sufficient supply of the food substance present in the solution **Q**……..
   (e) Give three symptoms of the deficiency you have mentioned
      (i)……..……
      (ii)……..……
      (iii)……

6. While in a biology session, students from Form Two class were told by their teacher that the scientific name for human being is *Homo sapiens*.
   (a) What system was used to determine this name? ……..……
   (b) Give six key points that should be observed when forming this name.
      (i)……..……
      (ii) ……..…….....
      (iii) ……..…….
      (iv)……..…….
      (v) ……..…….
      (vi) ……..

7. A 3-years old child has been brought to a dispensary following these complaints: He is very weak, cries and eats a lot and has persistent diarrhea. The child also looks skinny with extremely thin arms and legs, a swollen belly, wrinkled skin like an old person and hair looks yellowish and very thin.
   (i) What nutritional disorders is child suffering from?……
   (ii) What advice would you give to the child's parents?……
8. (a) Describes how a plants leaf is adapted to photosynthesis (five points)
   (b) Why is photosynthesis a vital process for all living things (five points)
9. In a biology practical lesson, the students observe a sunflower cell using a microscope. As one of the students who attended the lesson, draw the sunflower cells you observed and label the parts of the cell described in each of the following statements.
   (a) The semi-permeable membrane that encloses the cytoplasm of the cell.
   (b) A large cellular organelle that contains hereditary information.
   (c) The site for respiration in the cell.
   (d) A jelly–like mixture consisting mostly of water, located between cell membrane and nucleus.
   (e) A part that protects supports and gives shape to plant cells.

SECTION C (15 Marks) Answer question ten (10)

10. In a Kibesa village meeting, the villagers concluded that all insects are dangerous. They then decided to kill them by spraying insecticides. As a biologist and educated student on this matter, convince them with six points that insects are beneficial to their lives and should not be killed.